Q1). **Explain one technique each for Analog to Digital modulation and Digital to Analog modulation with the help of diagrams.**

ans. Analog to Digital modulation refers to the process of converting a continuous analog signal, such as human voice, into a digital form so that it can be processed, stored, or transmitted by digital systems. This process is essential in modern communication systems like telephony and audio recording.

One of the most commonly used techniques for this conversion is Pulse Code Modulation (PCM). PCM works in three important steps. The first step is sampling, where the analog signal is measured at regular intervals. According to the Nyquist theorem, the sampling rate must be at least twice the highest frequency present in the signal to avoid loss of information. The second step is quantization, where the sampled values are approximated to the nearest discrete level. This step introduces a small error known as quantization error. Finally, in the encoding step, these quantized values are converted into binary form for digital transmission.

Analog Signal ⟶ Sampler ⟶ Quantizer ⟶ Encoder ⟶ Digital Output

Another technique is Delta Modulation (DM), where instead of sending the full signal value, only the difference between successive samples is transmitted.

This method is simpler and requires less bandwidth, but it may introduce errors such as slope overload distortion.

On the other hand, Digital to Analog modulation is used when digital data needs to be transmitted over an analog medium, such as telephone lines. In this process, digital signals are converted into analog waveforms using a carrier signal.

There are three main techniques used in digital to analog modulation. In Amplitude Shift Keying (ASK), the amplitude of the carrier signal is varied according to the binary data. In Frequency Shift Keying (FSK), the frequency of the carrier changes depending on whether the bit is 0 or 1. In Phase Shift Keying (PSK), the phase of the signal is altered, which is more efficient and widely used in modern communication systems.

Digital Data $\longrightarrow$ Modulator $\longrightarrow$ Analog Signal

Thus, both analog-to-digital and digital-to-analog conversions play a crucial role in enabling communication between different types of systems.

Q2 a) **What are the Basic types of errors during data transmission, explain with the help of an example.**

b) **Find CRC for the data polynomial x8+x5+x3+x1+1 with the generator polynomial**

ans. Types of Errors in Data Transmission:

During data transmission, it is possible that the received data differs from the transmitted data due to various disturbances in the communication channel. Such differences are known as errors.

One common type of error is the single-bit error, in which only one bit in the data unit is altered during transmission. For example, if the transmitted data is 10101 and it is received as 10001, only one bit has changed. This type of error is relatively rare in real communication systems.

A more common type is the burst error, where multiple bits are affected within a short span. For instance, if the transmitted data is 10101101 and the received data is 10011101, several bits have changed. Burst errors are more likely to occur due to noise and interference in the channel.

Errors in data transmission are mainly caused by factors such as thermal noise, electromagnetic interference, and signal attenuation. Detecting and correcting these errors is an important part of data communication systems.

b) CRC Calculation

Cyclic Redundancy Check (CRC) is a widely used error detection technique in data communication. It works by treating the data as a polynomial and dividing it by a predefined generator polynomial.

To perform CRC, we first append zeros to the data equal to the degree of the generator polynomial, which is 4. Therefore, the new data becomes:

1001010110000

Next, we perform modulo-2 division of this binary data using the generator polynomial. In modulo-2 division, subtraction is replaced by XOR operation. The division is carried out step by step until the remainder is obtained.

The final remainder obtained after the division is 1110, which is the CRC code. This remainder is then appended to the original data to form the transmitted codeword:

1001010111110

At the receiver side, the same division is performed. If the remainder is zero, the data is considered error-free; otherwise, an error is detected.

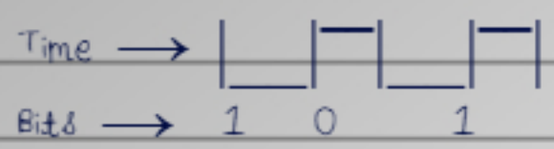
Q3). **Explain the concept of Digital-to-Digital encoding. Sketch the Differential Manchester coding and RZ encoding of the following bit stream:**

0011 0101

ans. <u>Digital-to-Digital Encoding</u>

Digital-to-digital encoding is the process of converting binary data into digital signals suitable for transmission over a communication channel. This process ensures proper synchronization between sender and receiver and helps in error detection.

One important encoding technique is Differential Manchester Encoding. In this method, there is always a transition at the middle of each bit interval, which helps in synchronization. The data is represented based on the presence or absence of a transition at the beginning of the bit period. If there is a transition at the beginning, it represents a binary 0; if there is no transition, it represents a binary 1.



This method has the advantage of being self-synchronizing and does not suffer from DC component issues.

Another encoding method is Return to Zero (RZ) encoding. In this scheme, the signal returns to zero in the middle of each bit interval. A binary 1 is represented by a positive pulse, while a binary 0 is represented by a negative pulse.

Although RZ encoding provides better synchronization, it requires more bandwidth compared to other encoding schemes.

Q4). a) Explain the ALOHA and Slotted ALOHA. Also, derive the expression for computing the throughput for ALOHA and Slotted ALOHA. How does throughout change with load in ALOHA? Explain with the help of a diagram.

b) Explain different versions of the CSMA protocol. Also, explain the CSMA/CD with the help of diagrams.

ALOHA and Slotted ALOHA

ans. ALOHA is a simple random access protocol used in communication networks where multiple users share a common channel. In pure ALOHA, a station transmits data whenever it has data to send, without checking whether the channel is free. This often leads to collisions when two or more stations transmit simultaneously.

$S = G e^{\wedge}\{-2G\}$

where S is the throughput and G is the offered load.

To improve efficiency, Slotted ALOHA was introduced. In this method, time is divided into discrete slots, and stations are allowed to transmit only at the beginning of a slot. This reduces the probability of collisions significantly.

$S = G e^{\wedge}\{-G\}$

Slotted ALOHA achieves a maximum efficiency of about 36%, which is double that of pure ALOHA.

b) CSMA Protocols

Carrier Sense Multiple Access (CSMA) is a protocol in which a station senses the channel before transmitting data. This helps in reducing collisions.

In 1-persistent CSMA, a station transmits immediately when the channel becomes idle. Although this reduces delay, it increases the chances of collision.

In non-persistent CSMA, if the channel is busy, the station waits for a random period before trying again. This reduces collisions but increases delay.

In p-persistent CSMA, the station transmits with a probability p when the channel is idle, providing a balance between delay and collision probability.

An advanced version of CSMA is CSMA/CD (Collision Detection), used in Ethernet networks. In this method, a station continues to monitor the channel even after transmission.

If a collision is detected, it immediately stops transmitting and retries after a random time.

Q5). **What is the purpose of Routing algorithms? Explain the link state routing algorithm with the help of an example.**

ans. Routing Algorithms

Routing algorithms are used to determine the best path for data to travel from the source to the destination in a network. These algorithms ensure efficient and reliable data delivery.

One important routing technique is the Link State Routing algorithm. In this method, each router gathers information about its neighbors and the cost of reaching them. This information is then shared with all other routers in the network.

Each router constructs a complete map of the network and uses Dijkstra's algorithm to compute the shortest path to every other node. This approach provides accurate and fast routing decisions.

Link state routing has advantages such as faster convergence and better performance compared to distance vector routing. However, it requires more memory and processing power.

Q6). **Explain the IEEE 802.11 protocol stack with the help of diagrams**

ans. IEEE 802.11 Protocol Stack

IEEE 802.11 is a standard used for wireless local area networks, commonly known as Wi-Fi. It defines the protocol stack required for wireless communication.

The protocol stack consists of multiple layers. The upper layers include the application, transport, and network layers, which are common to all network architectures.

Below these, the data link layer is divided into two sublayers: the Logical Link Control (LLC) and the Medium Access Control (MAC). The MAC layer is responsible for controlling access to the wireless medium and uses the CSMA/CA (Collision Avoidance) protocol.

At the lowest level is the physical layer, which handles the transmission of signals over the wireless medium. It defines parameters such as frequency bands and modulation techniques.

IEEE 802.11 supports mobility and allows devices to connect wirelessly within a limited range.

Q7). **Apply Dijkstra's shortest path algorithm to find the shortest path from a source node A to all the other nodes in a graph given below:**

ans. Dijkstra's algorithm is a greedy algorithm used to find the shortest path from a single source node to all other nodes in a graph with non-negative edge weights. It is widely used in routing protocols such as OSPF in computer networks. We are required to find the shortest path from node A to all other nodes in the given graph.

Step 1: Initialization

We begin by assigning:

Distance to source node A = 0

Distance to all other nodes = ∞ (infinity)

We also maintain a set of visited nodes.

Node	Distance	Previous Node
A	0	—
B	∞	—
C	∞	—
D	∞	—
E	∞	—
F	∞	—
G	∞	—

Step 2: Visit Node A

From the graph:

$A \longrightarrow B = 8$

$A \longrightarrow D = 5$

Update distances:

Node	Distance	Previous
A	0	——
B	8	A
D	5	A

Mark A as visited

Step 3: Visit Node with Minimum Distance $(D = 5)$

From node D:

$D \longrightarrow C = 5 \longrightarrow total = 5 + 5 = 10$

$D \longrightarrow F = 2 \longrightarrow total = 5 + 2 = 7$

Node	Distance	Previous
C	10	D
F	7	D

Mark D as visited

Step 4: Visit next minimum node $(F = 7)$

From node F:

$F \longrightarrow E = 7 \longrightarrow total = 7 + 7 = 14$

$F \longrightarrow G = 8 \longrightarrow total = 7 + 8 = 15$

Update:

Node	Distance	Previous
E	14	F
G	15	F

Mark F as visited

Step 5: Visit next minimum node $(B = 8)$

From node B:

$B \longrightarrow C = 4 \longrightarrow total = 8 + 4 = 12$ (no update, current is 10)

$B \longrightarrow E = 4 \longrightarrow total = 8 + 4 = 12$ (update, better than 14)

Update:

Node	Distance	Previous
E	12	B

Mark B as visited

Step 6: Visit Next Minimum Node $(C = 10)$

From node C:

No shorter path found

Mark C as visited

Step 7: Visit Next Minimum Node $(E = 12)$

From node E:

$E \longrightarrow G = 7 \longrightarrow total = 12 + 7 = 19$ (no update)

Mark E as visited

Step 8: Visit Last Node $(G = 15)$

Mark G as visited

Node	Distance	Path
A	0	A
B	8	A → B
C	10	A → D → C
D	5	A → D
E	12	A → B → E
F	7	A → D → F
G	15	A → D → F → G